

Vanessa Lai

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EDUCATION

McMaster University

Hamilton, ON

Bachelor of Engineering, Software Engineering

Sep. 2022 – Exp. May 2027

- **GPA: 3.8/4.0**, Motorola Software Engineering Scholarship, DELF B2 (Diploma in French Language Studies)
- **Courses:** Data Structures & Algorithms, Software Design & Requirements, Machine Learning, Statistics, Databases, Computer Architecture, Software Testing

EXPERIENCE

Royal Bank of Canada

Jan 2025 – Dec 2025

Software Engineer Intern

Toronto, ON

- Automated architecture diagrams from Dynatrace APIs, boosting developer productivity and saving **500+** hours
- Improved reliability for **50+** microservices, automated server monitoring and load testing for **15+** servers to help **50+** engineers reduce system downtime impacting **2000+** users
- Automated the configuration of non-production environments for platform tools (Devin AI) by developing a Python script, accelerating environment setup efficiency by **15%**
- Built custom GitHub Actions to visualize workflow runs and reduce debugging time by **15%** using OpenTelemetry and Dynatrace, for **20+** reusable workflows
- Led a proof-of-concept to integrate Checkmarx DAST Scans into CI/CD pipelines, improving application security

McMaster University

Sept 2024 – April 2026

Software Engineering Teaching Assistant

Hamilton, ON

- **Courses:** Object Oriented Programming (OOP), Computer Science Practice and Experience: Development Basics
- Collaborated with professors to develop course materials for labs and tutorials, automated the evaluation of student assignments making grading **44%** faster
- Mentored **400+** students Java, OOP, SOLID principles, UNIX/LINUX, Bash, C and Git

McMaster iGEM

April 2024 – Nov 2024

Front End Software Engineering Lead

Hamilton, ON

- Led a **5+** person team to develop of a silver medal research wiki at a conference with **5000+** participants
- Designed the project website using React and Tailwind CSS, enhancing user engagement by **35%** through responsive layouts, accessibility improvements, and consistent brand identity across pages

PROJECTS

Image Processor with Neural Network

Dec. 2024

- Created an image classifier using PyTorch, building and training a Convolutional Neural Network (CNN) model to classify images accurately, achieving **92.3%** classification accuracy while also implementing a 2-layer CNN denoiser to significantly improve image data quality
- Optimized the training pipeline across **73,000** images using normalized data transformations, demonstrating strong model generalization and efficiency improvements

BizConnet - Hackathon Winner | *Figma, JavaScript, Firebase*

Jan 2023

- Won **1st** Place RBC Sponsor Prize at DeltaHacks (**100+** teams across **20+** universities) for developing a matching algorithm that connects small businesses across Canada to exchange services
- Led frontend and UI/UX design development using Figma for high fidelity prototyping, and JavaScript, HTML/CSS, validating designs through user interviews and earning recognition for design quality

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, SQL, Bash, C++, JavaScript, HTML, CSS, Go, Verilog

Technologies: React, Node.js, Flask, JUnit, MongoDB, Supabase, Docker, PyTorch, TensorFlow, Kubernetes, Vue

Developer Tools: Git, Github, JIRA, Figma, GitHub Actions, JFrog, AWS, Maven, UNIX, Dynatrace, Ansible

Focus: CI/CD, Object Oriented Programming, Agile Methodologies, Rest API, Version Control, Scrum, DevOps, Site Reliability Engineering, DevOps, Front End, Full Stack, UI/UX